

# EQUIVARIANT COHOMOLOGICAL DESCENT AND THE WEIGHT FILTRATION

Let  $X$  be a projective complex variety, possibly singular and nonnormal. All cohomology groups have rational coefficients unless otherwise indicated and refer to the complex analytic spaces. We first construct the mixed Hodge structure on simplicial cohomology, then transport it to  $H^k(X, \mathbb{Q})$  by proper descent. Deligne's independence theorem makes the resulting structure canonical.

**Definition 1** (Hypercovector constructed by successive resolutions). We use the following construction of a smooth projective proper hypercover  $\pi_\bullet : X_\bullet \rightarrow X$ . Start with a projective resolution

$$\pi_0 : X_0 \longrightarrow X, \quad N_0 := X_0.$$

For a reducible space, take the disjoint union of resolutions of its reduced irreducible components; use this convention at every later stage as well.

Set  $Y_1 := X_0 \times_X X_0$ , and choose a projective resolution

$$r_1 : N_1 \longrightarrow (Y_1)_{\text{red}} \hookrightarrow Y_1, \quad X_1 := N_1 \sqcup X_0.$$

The face maps  $d_0^{(1)}, d_1^{(1)} : X_1 \rightarrow X_0$  restrict on  $N_1$  to the two projections composed with  $r_1$ , and on the extra  $X_0$  to the identity. The degeneracy map  $s_0^{(0)} : X_0 \rightarrow X_1$  includes that extra component. Set  $\pi_1 := \pi_0 d_0^{(1)} = \pi_0 d_1^{(1)}$ . The matching map  $X_1 \rightarrow Y_1$  is  $r_1$  on  $N_1$  and the diagonal on the extra  $X_0$ ; it is proper and surjective.

Suppose the levels below  $p \geq 2$  and their face and degeneracy maps have been constructed. Define their matching scheme by

$$Y_p := \left\{ (x_0, \dots, x_p) \in X_{p-1}^{\times_X (p+1)} : d_i^{(p-1)}(x_j) = d_{j-1}^{(p-1)}(x_i) \text{ for } i < j \right\}.$$

This is the scheme-theoretic equalizer of the displayed pairs of morphisms. Choose a projective resolution  $r_p : N_p \rightarrow (Y_p)_{\text{red}} \hookrightarrow Y_p$  and set

$$X_p := N_p \sqcup \coprod_{j=0}^{p-1} \coprod_{\eta: [p] \rightarrow [j]} N_j, \quad [p] := \{0, \dots, p\}.$$

Here  $\eta$  runs over order-preserving surjections; each copy of  $N_j$  is a degeneracy component labelled by  $\eta$ . For example,  $X_2 = N_2 \sqcup N_1 \sqcup N_1 \sqcup N_0$ . On  $N_p$ , define  $d_i^{(p)}$  by the  $i$ th projection of  $r_p$ . Deligne's split extension construction [D74, (6.2.4)–(6.2.5), pp. 19–20] extends the lower levels and these maps to  $X_p$  with the indicated degeneracy components. On the copy labelled by  $\eta : [p] \twoheadrightarrow [j]$ , a degeneracy map is the identity on  $N_j$  into the copy labelled by its composite with the corresponding surjection  $[p+1] \twoheadrightarrow [p]$ ; the remaining face maps are fixed by the lower-level maps and the simplicial identities.

The resulting maps

$$d_i^{(p)} : X_p \rightarrow X_{p-1} \quad (p \geq 1, 0 \leq i \leq p), \quad s_i^{(p)} : X_p \rightarrow X_{p+1} \quad (p \geq 0, 0 \leq i \leq p)$$

satisfy, with level superscripts omitted,

$$\begin{aligned} d_i d_j &= d_{j-1} d_i & (i < j), \\ s_i s_j &= s_{j+1} s_i & (i \leq j), \\ d_i s_j &= \begin{cases} s_{j-1} d_i, & i < j, \\ \text{id}, & i = j \text{ or } i = j + 1, \\ s_j d_{i-1}, & i > j + 1. \end{cases} \end{aligned}$$

Every identity is understood in the degrees where it is defined. The structure maps  $\pi_p : X_p \rightarrow X$  satisfy  $\pi_{p-1} d_i^{(p)} = \pi_p$  and  $\pi_{p+1} s_i^{(p)} = \pi_p$ .

Each  $Y_p$  is projective, so the chosen resolutions  $N_p$  and all the  $X_p$  are finite disjoint unions of smooth projective varieties. The matching maps

$$\mu_0 := \pi_0, \quad \mu_p := (d_0^{(p)}, \dots, d_p^{(p)}) : X_p \rightarrow Y_p \quad (p \geq 1)$$

are proper and surjective:  $\mu_0$  is the initial resolution, and for  $p \geq 1$  the restriction of  $\mu_p$  to  $N_p$  is the proper surjection  $r_p$ . This is Deligne's *proper hypercover* condition (*hyperrecouvrement propre*, [D74, (5.3.8)]). The symbol  $\bullet$  denotes all levels and structure maps.

Finally, if  $\pi_0 f_0 = f \pi_0$  for automorphisms  $f_0$  of  $X_0$  and  $f$  of  $X$ , choosing subsequent resolutions functorially under isomorphisms [BM08, Theorem 1.1] lifts the induced action on each  $Y_p$  to  $N_p$  and its degeneracy copies, yielding a simplicial automorphism  $f_\bullet = (f_p)_{p \geq 0}$  with

$$\pi_p f_p = f \pi_p, \quad d_i^{(p)} f_p = f_{p-1} d_i^{(p)}, \quad s_i^{(p)} f_p = f_{p+1} s_i^{(p)},$$

called a *lift of  $f$* , and the resulting hypercover is called  *$f$ -equivariant*.

**Definition 2.** For a hypercover as in Definition 1, put

$$K^{p,q} = C_{\text{sing}}^q(X_p^{\text{an}}, \mathbb{Q}) \quad (p, q \geq 0), \quad K^{p,q} = 0 \quad (p < 0 \text{ or } q < 0),$$

where  $C_{\text{sing}}^q$  is the space of singular cochains. Define the total cochain complex by

$$\begin{aligned} h|_{K^{p,q}} &= \sum_{i=0}^{p+1} (-1)^i (d_i^{(p+1)})^*, \quad v|_{K^{p,q}} = (-1)^p d_{\text{sing}}, \\ \text{Tot}^m K &= \bigoplus_{p+q=m} K^{p,q}, \quad D = h + v. \end{aligned}$$

The simplicial identities and naturality of  $d_{\text{sing}}$  give  $h^2 = v^2 = hv + vh = 0$ , hence  $D^2 = 0$ . Write

$$H^k(X_\bullet, \mathbb{Q}) := H^k(\text{Tot } K, D).$$

Equivalently, this is the cohomology of the simplicial analytic space, computed by  $R\Gamma(X_\bullet^{\text{an}}, \mathbb{Q})$ . When a lift  $f_\bullet$  is given, the levelwise pullbacks define

$$T^m = \bigoplus_{p=0}^m f_p^* : \text{Tot}^m K \longrightarrow \text{Tot}^m K, \quad (a_p)_p \longmapsto (f_p^* a_p)_p.$$

They satisfy  $DT^m = T^{m+1}D$ . Set  $f_\bullet^* = H^k(T)$  on  $H^k(X_\bullet, \mathbb{Q})$ . On the cohomology of the individual levels, define

$$\delta^{p,q} := \sum_{i=0}^{p+1} (-1)^i (d_i^{(p+1)})^* : H^q(X_p, \mathbb{Q}) \longrightarrow H^q(X_{p+1}, \mathbb{Q}).$$

Thus  $\delta^{p,q}([a]) = [ha]$  for a closed cochain  $a \in K^{p,q}$ . For  $p = 0$  we use  $\text{im } \delta^{-1,q} = 0$ .

The face identities and the equivariance of the face maps imply

$$\delta^{p+1,q}\delta^{p,q} = 0, \quad f_{p+1}^*\delta^{p,q} = \delta^{p,q}f_p^*.$$

Thus  $f_p^*$  induces an action on  $\ker \delta^{p,q} / \text{im } \delta^{p-1,q}$ .

**Definition 3.** Suppose each  $X_p$  is a finite disjoint union of smooth projective varieties. The filtrations on  $H^k(X_\bullet, \mathbb{Q})$  are defined as follows. On the rational total complex, set

$$L^p \text{Tot}^m K := \bigoplus_{i \geq p} K^{i, m-i},$$

and let

$$L^p H^k(\text{Tot } K, D) := \text{im} \left( H^k(L^p \text{Tot } K, D) \longrightarrow H^k(\text{Tot } K, D) \right).$$

The increasing weight filtration on simplicial cohomology is

$$W_w H^k(X_\bullet, \mathbb{Q}) := L^{k-w} H^k(\text{Tot } K, D).$$

To define the Hodge filtration, let  $\mathcal{A}^{a,b}(X_p)$  be the space of smooth complex-valued differential forms of type  $(a, b)$  on  $X_p^{\text{an}}$ . Put

$$B^m := \bigoplus_{p+a+b=m} \mathcal{A}^{a,b}(X_p), \quad D_B = h + (-1)^p d_{\text{dR}},$$

where all three indices are nonnegative and  $h$  is the alternating sum of face pullbacks. The degreewise de Rham comparisons give a natural identification

$$H^k(B^\bullet, D_B) \cong H^k(X_\bullet, \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C} =: H^k(X_\bullet, \mathbb{C}).$$

Set

$$F^\ell B^m := \bigoplus_{\substack{p+a+b=m \\ a \geq \ell}} \mathcal{A}^{a,b}(X_p).$$

Holomorphic pullbacks preserve type, while  $d_{\text{dR}} = \partial + \bar{\partial}$  does not decrease the first type index; hence  $F^\ell B^\bullet$  is a subcomplex. Under the comparison above, define

$$F^\ell H^k(X_\bullet, \mathbb{C}) := \text{im} \left( H^k(F^\ell B^\bullet, D_B) \longrightarrow H^k(B^\bullet, D_B) \right).$$

Thus  $W$  is a filtration over  $\mathbb{Q}$ , and  $F$  is a filtration on the complexification.

Deligne's construction [D74, (8.1.19)–(8.1.20)], applied with empty boundary in every smooth projective level, proves that

$$(H^k(X_\bullet, \mathbb{Q}), W, F)$$

is a mixed Hodge structure: the filtration induced by  $F$  on each  $\text{Gr}_w^W H^k(X_\bullet, \mathbb{Q})$  is a pure Hodge structure of weight  $w$ . The construction is natural for simplicial algebraic maps. In particular,  $f_\bullet^*$  is an automorphism of this mixed Hodge structure. These filtrations are constructed on  $X_\bullet$  before using the descent comparison with  $X$ .

To discuss equivariance, put  $n = p + q$  and start with an endomorphism  $u$  of the mixed Hodge structure  $H^n(X_\bullet, \mathbb{Q})$ . Since  $u$  preserves  $W$  and  $L^p H^n = W_{n-p} H^n$ , it induces

$$u_L^p := \text{Gr}_L^p(u), \quad u_L^p(\alpha + L^{p+1} H^n) = u(\alpha) + L^{p+1} H^n.$$

In our situation  $u = f_\bullet^* = H^n(T)$ , where  $T|_{K^{i,j}} = f_i^*$ . The simplicial identities for  $f_\bullet$  and naturality of cochain pullback give  $Th = hT$  and  $Tv = vT$ . Thus  $TD = DT$ ; in particular,  $T$  preserves  $L$  and induces the spectral-sequence endomorphisms  $T_r$ , with  $T_{r+1} = H(T_r)$ . The MHS condition on  $u$  supplies  $u_L^p$ ; its realization as  $H^n(T)$  is what relates this action to the given actions  $T_r$ .

The comparison map in the following lemma is the composite of two standard spectral-sequence identifications. The column filtration  $L$  gives

$$E_1^{p,q} = H^q(X_p, \mathbb{Q}), \quad E_2^{p,q} = \frac{\ker \delta^{p,q}}{\operatorname{im} \delta^{p-1,q}} \implies H^{p+q}(X_\bullet, \mathbb{Q}).$$

Since  $L$  is finite in each total degree, convergence gives the canonical isomorphism

$$\rho_{p,q} : \operatorname{Gr}_L^p H^{p+q}(X_\bullet, \mathbb{Q}) \xrightarrow{\sim} E_\infty^{p,q}.$$

It sends the class of a closed cochain  $a \in L^p \operatorname{Tot}^{p+q} K$ , modulo  $L^{p+1} H^{p+q}$ , to the stable spectral-sequence class represented by  $a$ . Write this stable class as  $[a]_\infty$ . Because  $u = H^n(T)$ , applying  $u$  replaces  $a$  by  $Ta$ ; by construction,  $T_\infty$  also sends  $[a]_\infty$  to  $[Ta]_\infty$ . Consequently,

$$\begin{aligned} \rho_{p,q} u_L^p([a]_D + L^{p+1} H^n) &= [Ta]_\infty \\ &= T_\infty^{p,q} \rho_{p,q}([a]_D + L^{p+1} H^n), \quad \rho_{p,q} u_L^p = T_\infty^{p,q} \rho_{p,q}. \end{aligned}$$

This proves the equivariance of  $\rho$  directly on representatives.

Separately, degeneration at  $E_2$  in the smooth projective setting, proved below, means that  $D_r = 0$  for all  $r \geq 2$ . Thus  $E_{r+1} = \ker D_r / \operatorname{im} D_r$  is canonically identified with  $E_r$  for every  $r \geq 2$ , giving

$$\sigma_{p,q} : E_2^{p,q} \xrightarrow{\sim} E_\infty^{p,q}.$$

The relation  $T_{r+1} = H(T_r)$  means that each of these identifications intertwines  $T_r$  and  $T_{r+1}$ . Hence

$$\sigma_{p,q} T_2^{p,q} = T_\infty^{p,q} \sigma_{p,q}.$$

The desired map is therefore

$$\beta_{p,q} := \sigma_{p,q}^{-1} \circ \rho_{p,q} : \operatorname{Gr}_L^p H^{p+q}(X_\bullet, \mathbb{Q}) \xrightarrow[\rho_{p,q}]{\sim} E_\infty^{p,q} \xrightarrow[\sigma_{p,q}^{-1}]{\sim} E_2^{p,q}.$$

Here  $\rho$  comes from convergence, while  $\sigma$  comes from  $E_2$ -degeneration; convergence alone does not supply the inverse arrow from  $E_\infty$  to  $E_2$ . Their equivariance gives

$$\beta_{p,q} u_L^p = \sigma_{p,q}^{-1} T_\infty^{p,q} \rho_{p,q} = T_2^{p,q} \beta_{p,q}.$$

On  $E_2^{p,q} = \ker \delta^{p,q} / \operatorname{im} \delta^{p-1,q}$ , the action is  $T_2^{p,q}([\alpha]) = [f_p^* \alpha]$ .

Explicitly, write the closed representative as  $a = \sum_{i=p}^{p+q} a_i$ , with  $a_i \in K^{i,p+q-i}$ . The equations  $va_p = 0$  and  $ha_p + va_{p+1} = 0$  (with  $a_{p+1} = 0$  when  $q = 0$ ) show that  $[a_p]_{\operatorname{sing}} \in \ker \delta^{p,q}$ , and

$$\beta_{p,q}([a]_D + L^{p+1} H^{p+q}(X_\bullet, \mathbb{Q})) = [a_p]_{\operatorname{sing}} + \operatorname{im} \delta^{p-1,q}.$$

Here  $[a]_D$  and  $[a_p]_{\operatorname{sing}}$  denote total and singular cohomology classes, respectively. The construction through  $\rho$  and  $\sigma$  ensures independence of the representative.

**Lemma 1** (Column-graded cohomology as a subquotient). *Let  $X_\bullet$  be a smooth projective simplicial scheme, equipped with the filtrations of Definition 3. For  $p, q \geq 0$  there is a canonical isomorphism of pure Hodge structures of weight  $q$ ,*

$$\beta_{p,q} : \operatorname{Gr}_L^p H^{p+q}(X_\bullet, \mathbb{Q}) \xrightarrow{\sim} \frac{\ker \delta^{p,q}}{\operatorname{im} \delta^{p-1,q}}.$$

For any simplicial automorphism  $f_\bullet$ , the following square commutes, with the right vertical map induced by  $f_p^*$ :

$$\begin{array}{ccc} \mathrm{Gr}_L^p H^{p+q}(X_\bullet, \mathbb{Q}) & \xrightarrow[\sim]{\beta_{p,q}} & \frac{\ker \delta^{p,q}}{\mathrm{im} \delta^{p-1,q}} \\ \downarrow \mathrm{Gr}_L^p(f_\bullet^*) & & \downarrow f_p^* \\ \mathrm{Gr}_L^p H^{p+q}(X_\bullet, \mathbb{Q}) & \xrightarrow[\sim]{\beta_{p,q}} & \frac{\ker \delta^{p,q}}{\mathrm{im} \delta^{p-1,q}}. \end{array}$$

*Proof.* The column filtration  $L$  of Definition 3 gives the first-quadrant spectral sequence

$$\begin{aligned} E_1^{p,q} &= H^q(X_p, \mathbb{Q}), \quad D_1^{p,q} = \delta^{p,q}, \\ E_2^{p,q} &= \frac{\ker \delta^{p,q}}{\mathrm{im} \delta^{p-1,q}}, \quad E_1^{p,q} \implies H^{p+q}(\mathrm{Tot} K, D). \end{aligned}$$

For the asserted compatibility,  $T|_{K^{p,q}} = f_p^*$  preserves bidegrees. Compatibility with the face maps gives  $Th = hT$ , and naturality of the singular cochain differential gives  $Tv = vT$ ; hence  $TD = DT$ . In particular,  $T$  preserves the column filtration  $L$  and is an endomorphism of the filtered complex. It induces maps  $T_r^{p,q} : E_r^{p,q} \rightarrow E_r^{p,q}$  satisfying

$$D_r^{p,q} T_r^{p,q} = T_r^{p+r,q-r+1} D_r^{p,q}, \quad T_{r+1} = H(T_r)$$

[Stacks, Lemma 12.24.4]. On the first two pages these maps are

$$T_1^{p,q} = f_p^*, \quad T_2^{p,q}(\alpha + \mathrm{im} \delta^{p-1,q}) = f_p^* \alpha + \mathrm{im} \delta^{p-1,q} \quad (\alpha \in \ker \delta^{p,q}).$$

The second formula is well-defined by  $f_{p+1}^* \delta^{p,q} = \delta^{p,q} f_p^*$  and, for  $p \geq 1$ ,  $f_p^* \delta^{p-1,q} = \delta^{p-1,q} f_{p-1}^*$ ; when  $p = 0$  the incoming image is zero.

Since  $X_p$  is smooth projective,  $E_1^{p,q}$  is pure of weight  $q$ . The maps  $\delta^{p,q}$  are morphisms of Hodge structures, so their kernel modulo image  $E_2^{p,q}$  is pure of weight  $q$ . Deligne's theory makes every differential

$$D_r^{p,q} : E_r^{p,q} \longrightarrow E_r^{p+r,q-r+1}$$

a morphism of Hodge structures [D74, Proposition (8.3.5)]. For  $r \geq 2$ , induction gives pure weights  $q$  and  $q - r + 1$  on its source and target. These weights differ, so  $D_r^{p,q} = 0$ . Thus the canonical identifications  $E_{r+1}^{p,q} = H^{p,q}(E_r, D_r) \cong E_r^{p,q}$  for  $r \geq 2$  give the isomorphism  $\sigma_{p,q} : E_2^{p,q} \xrightarrow{\sim} E_\infty^{p,q}$ . The relation  $T_{r+1} = H(T_r)$  gives the square

$$\begin{array}{ccc} E_2^{p,q} & \xrightarrow[\sim]{\sigma_{p,q}} & E_\infty^{p,q} \\ T_2^{p,q} \downarrow & & \downarrow T_\infty^{p,q} \\ E_2^{p,q} & \xrightarrow[\sim]{\sigma_{p,q}} & E_\infty^{p,q}. \end{array}$$

Convergence for the degreewise finite filtration  $L$  [Stacks, Lemma 12.24.11] supplies

$$\rho_{p,q} : \mathrm{Gr}_L^p H^{p+q}(X_\bullet, \mathbb{Q}) \xrightarrow{\sim} E_\infty^{p,q}.$$

Explicitly, a class in  $L^p H^{p+q}$  has a closed representative  $a \in L^p \mathrm{Tot}^{p+q} K$ . The map  $\rho_{p,q}$  sends its class modulo  $L^{p+1} H^{p+q}$  to the stable spectral-sequence class represented by  $a$ . The well-definedness and bijectivity are the standard convergence identification. Since

$T$  sends this representative to  $Ta$ , and  $f_\bullet^* = H^{p+q}(T)$ , both paths below send the class of  $a$  to the stable class of  $Ta$ :

$$\begin{array}{ccc} \mathrm{Gr}_L^p H^{p+q}(X_\bullet, \mathbb{Q}) & \xrightarrow[\sim]{\rho_{p,q}} & E_\infty^{p,q} \\ \mathrm{Gr}_L^p(f_\bullet^*) \downarrow & & \downarrow T_\infty^{p,q} \\ \mathrm{Gr}_L^p H^{p+q}(X_\bullet, \mathbb{Q}) & \xrightarrow[\sim]{\rho_{p,q}} & E_\infty^{p,q}. \end{array}$$

Deligne's compatibility of the spectral sequence with the weight filtration identifies  $\mathrm{Gr}_L^p H^{p+q}$  with  $\mathrm{Gr}_q^W H^{p+q}$  and makes  $\rho_{p,q}$  and  $\sigma_{p,q}$  isomorphisms of pure Hodge structures of weight  $q$  [D74, (8.1.20)(i)–(ii)].

The comparison introduced above is

$$\beta_{p,q} := \sigma_{p,q}^{-1} \rho_{p,q} : \mathrm{Gr}_L^p H^{p+q}(X_\bullet, \mathbb{Q}) \xrightarrow{\sim} E_2^{p,q} = \frac{\ker \delta^{p,q}}{\mathrm{im} \delta^{p-1,q}}.$$

For the closed representative  $a$  used above,  $\rho_{p,q}$  gives its stable class, while  $\sigma_{p,q}$  sends the  $E_2$ -class of  $[a_p]_{\mathrm{sing}}$  to that same stable class. Thus this composite has the stated representative formula and proves its independence of choices. The two squares give

$$\begin{aligned} \beta_{p,q} \mathrm{Gr}_L^p(f_\bullet^*) &= \sigma_{p,q}^{-1} T_\infty^{p,q} \rho_{p,q} \\ &= T_2^{p,q} \beta_{p,q}. \end{aligned}$$

The formula for  $T_2^{p,q}$  proves the displayed equivariance.  $\square$

Proper cohomological descent gives, for a proper smooth projective hypercover  $\pi_\bullet : X_\bullet \rightarrow X$ , a natural isomorphism of rational vector spaces

$$\pi_\bullet^* : H^k(X, \mathbb{Q}) \xrightarrow{\sim} H^k(X_\bullet, \mathbb{Q}) \quad (k \geq 0)$$

[D74, (8.2.1.2), p. 38]. This comparison is established before putting a mixed Hodge structure on  $H^k(X, \mathbb{Q})$ .

**Definition 4** (Mixed Hodge structure transported to  $X$ ). Choose a proper smooth projective hypercover  $\pi_\bullet : X_\bullet \rightarrow X$  and equip its cohomology with the filtrations of Definition 3. Define

$$\begin{aligned} W_w H^k(X, \mathbb{Q}) &:= (\pi_\bullet^*)^{-1} (W_w H^k(X_\bullet, \mathbb{Q})), \\ F^\ell H^k(X, \mathbb{C}) &:= (\pi_\bullet^* \otimes \mathbb{C})^{-1} (F^\ell H^k(X_\bullet, \mathbb{C})). \end{aligned}$$

These filtrations make  $H^k(X, \mathbb{Q})$  a mixed Hodge structure by transport of structure. For this chosen hypercover,  $\pi_\bullet^*$  is an isomorphism of mixed Hodge structures by definition. At this stage the construction uses the choice of  $\pi_\bullet$ ; independence of that choice is the content of the following theorem.

**Theorem 1** (Deligne: independence and naturality). *The mixed Hodge structure of Definition 4 is independent of the chosen proper smooth projective hypercover. It is called Deligne's canonical mixed Hodge structure on  $H^k(X, \mathbb{Q})$ , and is functorial for algebraic morphisms of projective complex varieties. Consequently, for every proper smooth projective hypercover  $\pi_\bullet : X_\bullet \rightarrow X$ , the augmentation induces an isomorphism of mixed Hodge structures*

$$\pi_\bullet^* : H^k(X, \mathbb{Q}) \xrightarrow{\sim} H^k(X_\bullet, \mathbb{Q}).$$

For every integer  $w$ , its induced map on the weight-graded pieces, using  $W_w = L^{k-w}$  on the target as in Definition 3, is the isomorphism

$$\begin{aligned} \gamma_{k,w} : \mathrm{Gr}_w^W H^k(X, \mathbb{Q}) &\xrightarrow{\sim} \mathrm{Gr}_L^{k-w} H^k(X_\bullet, \mathbb{Q}), \\ \alpha + W_{w-1} H^k(X, \mathbb{Q}) &\mapsto \pi_\bullet^* \alpha + L^{k-w+1} H^k(X_\bullet, \mathbb{Q}). \end{aligned}$$

If the hypercover is equipped with a lift  $f_\bullet$  of  $f$ , then  $f_\bullet^* \pi_\bullet^* = \pi_\bullet^* f^*$ , and the induced square commutes:

$$\begin{array}{ccc} \mathrm{Gr}_w^W H^k(X, \mathbb{Q}) & \xrightarrow[\sim]{\gamma_{k,w}} & \mathrm{Gr}_L^{k-w} H^k(X_\bullet, \mathbb{Q}) \\ \mathrm{Gr}_w^W(f^*) \downarrow & & \downarrow \mathrm{Gr}_L^{k-w}(f_\bullet^*) \\ \mathrm{Gr}_w^W H^k(X, \mathbb{Q}) & \xrightarrow[\sim]{\gamma_{k,w}} & \mathrm{Gr}_L^{k-w} H^k(X_\bullet, \mathbb{Q}). \end{array}$$

*Proof.* Independence of the hypercover and functoriality are Deligne's Proposition (8.2.2), following the transported construction in (8.2.1) [D74, p. 38]. For any given hypercover, its transported filtrations therefore equal the canonical ones on  $H^k(X, \mathbb{Q})$ ; Definition 4 then makes its augmentation an isomorphism of mixed Hodge structures. Taking the quotient of  $W_w$  by  $W_{w-1}$  gives  $\gamma_{k,w}$ , since these subspaces on the target are  $L^{k-w}$  and  $L^{k-w+1}$ , respectively. Naturality applied to  $\pi_\bullet f_\bullet = f \pi_\bullet$  gives  $f_\bullet^* \pi_\bullet^* = \pi_\bullet^* f^*$ . For  $\alpha \in W_w H^k(X, \mathbb{Q})$ , the two paths in the square send its class to the classes of  $\pi_\bullet^* f^* \alpha$  and  $f_\bullet^* \pi_\bullet^* \alpha$ , respectively. They are equal.  $\square$

**Theorem 2.** Let  $\pi_\bullet : X_\bullet \rightarrow X$  be an  $f$ -equivariant proper smooth projective hypercover. For every  $k \geq 0$  and  $0 \leq w \leq k$ , there is a natural isomorphism of pure Hodge structures of weight  $w$ ,

$$\theta_{k,w} : \mathrm{Gr}_w^W H^k(X, \mathbb{Q}) \xrightarrow{\sim} \frac{\ker \delta^{k-w,w}}{\mathrm{im} \delta^{k-w-1,w}},$$

with the following compatibility:

$$\begin{array}{ccc} \mathrm{Gr}_w^W H^k(X, \mathbb{Q}) & \xrightarrow[\sim]{\theta_{k,w}} & \frac{\ker \delta^{k-w,w}}{\mathrm{im} \delta^{k-w-1,w}} \\ \mathrm{Gr}_w^W(f^*) \downarrow & & \downarrow f_{k-w}^* \\ \mathrm{Gr}_w^W H^k(X, \mathbb{Q}) & \xrightarrow[\sim]{\theta_{k,w}} & \frac{\ker \delta^{k-w,w}}{\mathrm{im} \delta^{k-w-1,w}}. \end{array}$$

The right vertical map sends the class of  $\alpha$  to the class of  $f_{k-w}^* \alpha$ . For  $w < 0$  or  $w > k$ ,  $\mathrm{Gr}_w^W H^k(X, \mathbb{Q}) = 0$ .

*Proof.* Compose the isomorphisms of Theorem 1 and Lemma 1:

$$\begin{aligned} \theta_{k,w} := \beta_{k-w,w} \gamma_{k,w} : \quad & \mathrm{Gr}_w^W H^k(X, \mathbb{Q}) \xrightarrow{\gamma_{k,w}} \mathrm{Gr}_L^{k-w} H^k(X_\bullet, \mathbb{Q}) \\ & \xrightarrow{\beta_{k-w,w}} \frac{\ker \delta^{k-w,w}}{\mathrm{im} \delta^{k-w-1,w}}. \end{aligned}$$

Both arrows are isomorphisms of pure Hodge structures of weight  $w$ . Their commutative squares give

$$\begin{aligned} \theta_{k,w} \mathrm{Gr}_w^W(f^*) &= \beta_{k-w,w} \mathrm{Gr}_L^{k-w}(f_\bullet^*) \gamma_{k,w} \\ &= f_{k-w}^* \beta_{k-w,w} \gamma_{k,w} = f_{k-w}^* \theta_{k,w}, \end{aligned}$$

where  $f_{k-w}^*$  on the last line denotes the quotient action. Finally, if  $w < 0$  or  $w > k$ , the first-quadrant condition and convergence give  $\mathrm{Gr}_L^{k-w} H^k(X_\bullet, \mathbb{Q}) \cong E_\infty^{k-w,w} = 0$ ;  $\gamma_{k,w}$  then gives  $\mathrm{Gr}_w^W H^k(X, \mathbb{Q}) = 0$ .  $\square$

## REFERENCES

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